🔋 Electric Vehicle Sales in India - Exploratory Data Analysis

📌 Objective:

To analyze the growth and distribution of Electric Vehicle (EV) sales across different Indian states by vehicle types, categories, and time. The goal is to identify trends and derive actionable insights that can support infrastructure, investment, and policy decisions.

Tools Used: Python, Pandas, Seaborn, Matplotlib

Dataset Source: Clean Mobility Shift (scraped + preprocessed)

In [1]:
```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

sns.set(style='whitegrid')
plt.rcParams['figure.figsize'] = (12, 6)
```

In [3]:
```
df = pd.read_csv(r'C:\Python Project\Electric Vehicles Sales\Electric Vehicle Sales by State in India.csv')
df.head(10)
```

Out[3]:

	Year	Month_Name	Date	State	Vehicle_Class	Vehicle_Category	Vehicle_Type	EV_Sales_Quantity
0	2014.0	jan	1/1/2014	Andhra Pradesh	ADAPTED VEHICLE	Others	Others	0.0
1	2014.0	jan	1/1/2014	Andhra Pradesh	AGRICULTURAL TRACTOR	Others	Others	0.0
2	2014.0	jan	1/1/2014	Andhra Pradesh	AMBULANCE	Others	Others	0.0
3	2014.0	jan	1/1/2014	Andhra Pradesh	ARTICULATED VEHICLE	Others	Others	0.0
4	2014.0	jan	1/1/2014	Andhra Pradesh	BUS	Bus	Bus	0.0
5	2014.0	jan	1/1/2014	Andhra Pradesh	CASH VAN	Others	Others	0.0
6	2014.0	jan	1/1/2014	Andhra Pradesh	CRANE MOUNTED VEHICLE	Others	Others	0.0
7	2014.0	jan	1/1/2014	Andhra Pradesh	EDUCATIONAL INSTITUTION BUS	Bus	Institution Bus	0.0
8	2014.0	jan	1/1/2014	Andhra Pradesh	EXCAVATOR (COMMERCIAL)	Others	Others	0.0
9	2014.0	jan	1/1/2014	Andhra Pradesh	FORK LIFT	Others	Others	0.0

In [3]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 96845 entries, 0 to 96844
Data columns (total 8 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Year               96845 non-null  float64
 1   Month_Name         96845 non-null  object
 2   Date               96845 non-null  object
 3   State              96845 non-null  object
 4   Vehicle_Class      96845 non-null  object
 5   Vehicle_Category   96845 non-null  object
 6   Vehicle_Type       96845 non-null  object
 7   EV_Sales_Quantity  96845 non-null  float64
dtypes: float64(2), object(6)
memory usage: 5.9+ MB
```

In [4]:
```
# Year type
df['Year'] = df['Year'].astype(int)

# Date to datetime
df['Date'] = pd.to_datetime(df['Date'], errors='coerce')

# categories
cat_cols = ['Month_Name', 'State', 'Vehicle_Class', 'Vehicle_Category', 'Vehicle_Type']
df[cat_cols] = df[cat_cols].astype('category')

# shape
print(f"Rows: {df.shape[0]}, Columns: {df.shape[1]}")
df.info()
```

```
Rows: 96845, Columns: 8
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 96845 entries, 0 to 96844
Data columns (total 8 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Year               96845 non-null  int64
 1   Month_Name         96845 non-null  category
 2   Date               96845 non-null  datetime64[ns]
 3   State              96845 non-null  category
 4   Vehicle_Class      96845 non-null  category
 5   Vehicle_Category   96845 non-null  category
 6   Vehicle_Type       96845 non-null  category
 7   EV_Sales_Quantity  96845 non-null  float64
dtypes: category(5), datetime64[ns](1), float64(1), int64(1)
memory usage: 2.7 MB
```

In [5]:
```
# missing values
print("Missing values:\n", df.isnull().sum())

# duplicates
print("\nDuplicate rows:", df.duplicated().sum())
```

```
Missing values:
 Year                0
Month_Name          0
Date                0
State               0
Vehicle_Class       0
Vehicle_Category    0
Vehicle_Type        0
EV_Sales_Quantity   0
dtype: int64

Duplicate rows: 0
```

In [11]:
```
state_year = df.groupby(['State', 'Year'])['EV_Sales_Quantity'].sum().unstack().fillna(0)
growth = state_year.pct_change(axis=1) * 100

print(state_year)
```

```
Year                      2014.0  2015.0   2016.0   2017.0   2018.0   2019.0  \
State
Andaman & Nicobar Island     0.0     0.0      0.0      0.0     20.0      2.0
Andhra Pradesh              12.0     8.0     20.0      0.0   1165.0   2123.0
Arunachal Pradesh            0.0     0.0      1.0      0.0      1.0      1.0
Assam                        2.0    23.0     19.0   3833.0   7403.0  10867.0
Bihar                       12.0    69.0   1100.0   3593.0   8469.0  12380.0
Chandigarh                   0.0     2.0      1.0     23.0    213.0    506.0
Chhattisgarh               123.0    86.0    192.0    945.0   1902.0   2744.0
DNH and DD                   0.0     2.0      2.0      1.0     19.0     19.0
Delhi                       48.0  3565.0  21688.0  18042.0  20961.0  23222.0
Goa                          9.0     4.0     34.0     39.0     26.0     41.0
Gujarat                     76.0    81.0    142.0    174.0    498.0    945.0
Haryana                      9.0    55.0   1381.0   2488.0   4623.0   5108.0
Himachal Pradesh             1.0     4.0      4.0     15.0     62.0     54.0
Jammu and Kashmir            4.0     5.0      0.0      1.0     90.0     76.0
Jharkhand                    4.0   194.0    705.0   1352.0   1957.0   1931.0
Karnataka                  601.0   682.0    605.0    729.0   2302.0   6152.0
Kerala                      26.0    27.0     19.0     77.0    264.0    471.0
Ladakh                       0.0     0.0      0.0      0.0      0.0      0.0
Madhya Pradesh              41.0    84.0    836.0   1404.0   2738.0   3680.0
Maharashtra                905.0  1015.0    951.0    945.0   4640.0   7319.0
Manipur                      0.0     0.0      2.0      5.0     39.0    251.0
Meghalaya                    1.0     0.0      1.0      0.0      2.0     13.0
Mizoram                      0.0     0.0      2.0      0.0      0.0      1.0
Nagaland                     0.0     1.0      0.0      0.0      0.0     29.0
Odisha                      51.0    47.0     25.0    212.0   1358.0   1159.0
Puducherry                   3.0    11.0      7.0      9.0     87.0     80.0
Punjab                      30.0    19.0     38.0    139.0    363.0    961.0
Rajasthan                   87.0   676.0   3996.0   3945.0   4678.0   6633.0
Sikkim                       1.0     2.0      1.0      0.0      0.0      0.0
Tamil Nadu                 196.0    86.0     86.0    118.0   1332.0   3444.0
Tripura                      0.0     0.0      0.0      2.0     32.0   1459.0
Uttar Pradesh              139.0  1012.0  15310.0  40649.0  53212.0  55796.0
Uttarakhand                  9.0    14.0   1158.0   4225.0   5070.0   5819.0
West Bengal                  2.0    31.0   1529.0   4455.0   6728.0  13533.0

Year                       2020.0   2021.0    2022.0    2023.0   2024.0
State
Andaman & Nicobar Island     36.0     92.0      23.0      26.0      3.0
Andhra Pradesh             1623.0   9540.0   29006.0   31506.0   2353.0
Arunachal Pradesh             5.0      2.0       2.0      21.0      7.0
Assam                      8357.0  15633.0   40720.0   60798.0   4262.0
Bihar                     12447.0  23082.0   55752.0   88186.0   8375.0
Chandigarh                  369.0    734.0    2719.0    6333.0    553.0
Chhattisgarh               1489.0   4214.0   22363.0   37638.0   3579.0
DNH and DD                   24.0     29.0     140.0     160.0     35.0
Delhi                     12378.0  25815.0   62265.0   73487.0   7067.0
Goa                          80.0   1095.0    5686.0    9440.0    928.0
Gujarat                    1120.0   9775.0   68997.0   88617.0   6288.0
Haryana                    2982.0   8661.0   25859.0   30489.0   3595.0
Himachal Pradesh            181.0    327.0    1008.0    1126.0    104.0
Jammu and Kashmir            74.0   1146.0    4690.0    9720.0   1034.0
Jharkhand                  1516.0   3741.0   13681.0   21097.0   1693.0
Karnataka                  9716.0  33314.0   95934.0  151272.0  14191.0
Kerala                     1386.0   8801.0   39671.0   74714.0   7790.0
Ladakh                        0.0      6.0      40.0      28.0     14.0
Madhya Pradesh             3358.0  10427.0   36797.0   67051.0   6766.0
Maharashtra                7137.0  29907.0  136036.0  193935.0  18745.0
Manipur                     104.0    114.0     341.0     399.0     29.0
Meghalaya                     3.0      5.0      42.0     113.0     28.0
Mizoram                       1.0      0.0      36.0     166.0     29.0
Nagaland                     11.0      2.0       3.0       6.0      0.0
Odisha                      904.0   5627.0   28444.0   44552.0   4444.0
Puducherry                   88.0    405.0    1481.0    2637.0    331.0
Punjab                      832.0   4643.0   14054.0   25741.0   2465.0
Rajasthan                  5603.0  23462.0   78240.0   93818.0   7435.0
Sikkim                        0.0      0.0       0.0       0.0      0.0
Tamil Nadu                 5698.0  30030.0   66955.0   90242.0   8715.0
Tripura                    3421.0   2435.0    4178.0    6312.0    507.0
Uttar Pradesh             31264.0  66702.0  162857.0  277300.0  27833.0
Uttarakhand                2398.0   5324.0   15561.0   16794.0   1234.0
West Bengal               10079.0   6408.0   11142.0   21455.0   2750.0
```

In [12]:
```
df['Cumulative_Sales'] = df.sort_values('Date').groupby('State')['EV_Sales_Quantity'].cumsum()
```

In [14]:
```
df[df['State'] == 'Maharashtra'][['Date', 'EV_Sales_Quantity', 'Cumulative_Sales']].sort_values('Date').head(15)
```

Out[14]:

	Date	EV_Sales_Quantity	Cumulative_Sales
348	1/1/2014	0.0	85.0
371	1/1/2014	0.0	82.0
372	1/1/2014	3.0	85.0
373	1/1/2014	0.0	85.0
374	1/1/2014	0.0	85.0
375	1/1/2014	0.0	85.0
376	1/1/2014	0.0	85.0
377	1/1/2014	0.0	85.0
379	1/1/2014	0.0	0.0
380	1/1/2014	0.0	0.0
381	1/1/2014	0.0	0.0
382	1/1/2014	0.0	0.0
383	1/1/2014	0.0	0.0
384	1/1/2014	0.0	0.0
385	1/1/2014	0.0	0.0

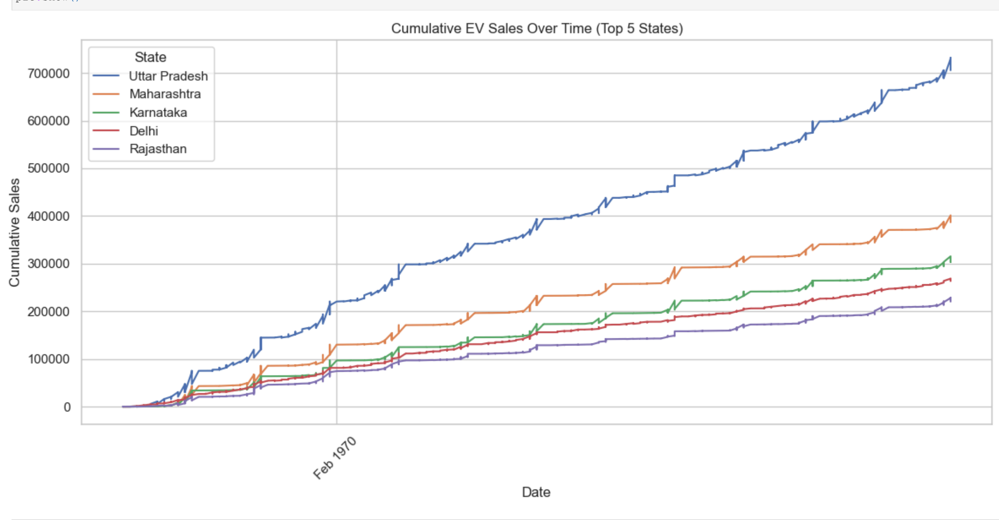
In [16]:
```
import matplotlib.dates as mdates

# Plot cumulative sales
plt.figure(figsize=(12, 6))

for state in top_states:
    state_data = df_top[df_top['State'] == state].sort_values('Date')
    plt.plot(state_data['Date'], state_data['Cumulative_Sales'], label=state)

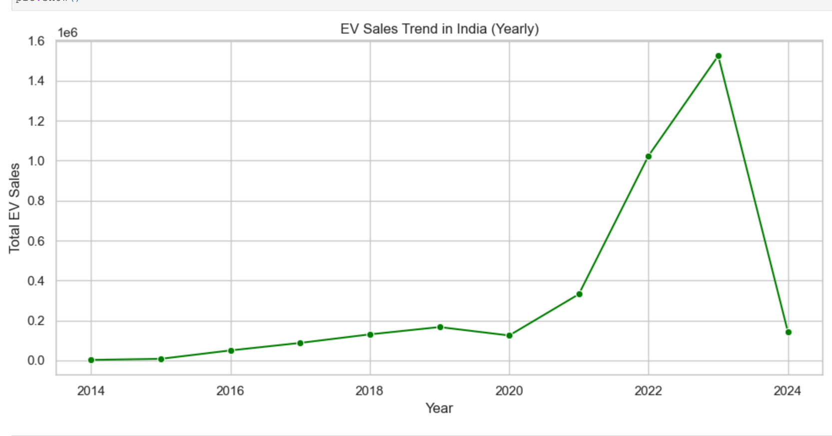
# Format the x-axis dates
plt.gca().xaxis.set_major_formatter(mdates.DateFormatter('%b %Y'))  # e.g., Jan 2021
plt.gca().xaxis.set_major_locator(mdates.MonthLocator(interval=6))  # every 6 months

plt.title('Cumulative EV Sales Over Time (Top 5 States)')
plt.xlabel('Date')
plt.ylabel('Cumulative Sales')
plt.legend(title='State')
plt.grid(True)
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



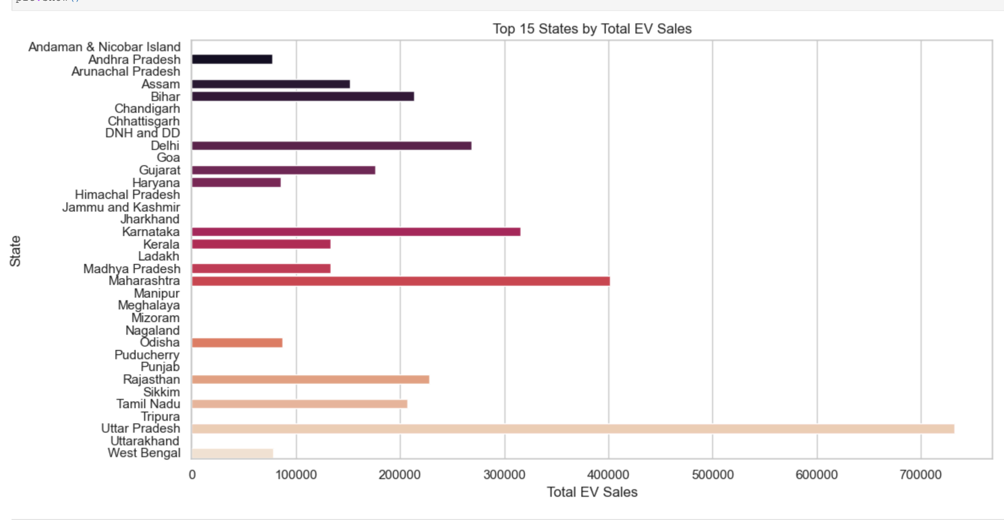
In [6]:
```
# yearly sum of EV sales
yearly_sales = df.groupby('Year')['EV_Sales_Quantity'].sum().reset_index()

plt.figure(figsize=(10, 5))
sns.lineplot(data=yearly_sales, x='Year', y='EV_Sales_Quantity', marker='o', color='green')
plt.title('EV Sales Trend in India (Yearly)')
plt.ylabel('Total EV Sales')
plt.xlabel('Year')
plt.grid(True)
plt.tight_layout()
plt.show()
```



In [7]:
```
# state-wise total EV sales
top_states = df.groupby('State', observed=False)['EV_Sales_Quantity'].sum().sort_values(ascending=False).head(15)

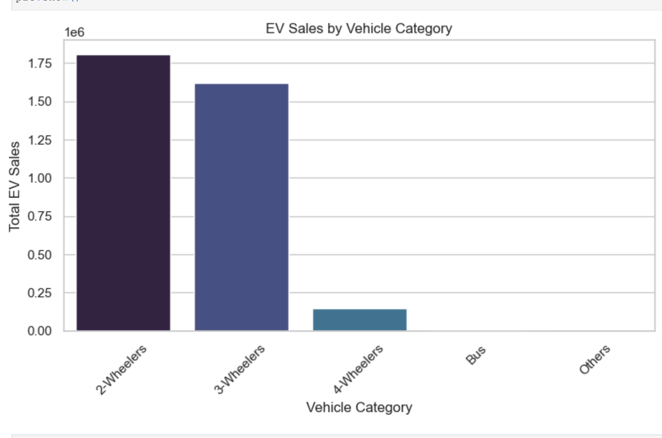
top_states_df = top_states.reset_index()
plt.figure(figsize=(12, 6))
sns.barplot(data=top_states_df, x='EV_Sales_Quantity', y='State', hue='State', dodge=False, palette='rocket', legend=False)
plt.title('Top 15 States by Total EV Sales')
plt.xlabel('Total EV Sales')
plt.ylabel('State')
plt.tight_layout()
plt.show()
```



In [15]:
```
# total sales
category_sales = df.groupby('Vehicle_Category', observed=True)['EV_Sales_Quantity'].sum().sort_values(ascending=False)

category_sales_df = category_sales.reset_index()
category_sales_df.columns = ['Vehicle_Category', 'EV_Sales_Quantity']

plt.figure(figsize=(8, 5))
sns.barplot(
    data=category_sales_df,
    x='Vehicle_Category',
    y='EV_Sales_Quantity',
    hue='Vehicle_Category',
    dodge=False,
    palette='mako',
    legend=False
)
plt.title('EV Sales by Vehicle Category')
plt.ylabel('Total EV Sales')
plt.xlabel('Vehicle Category')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

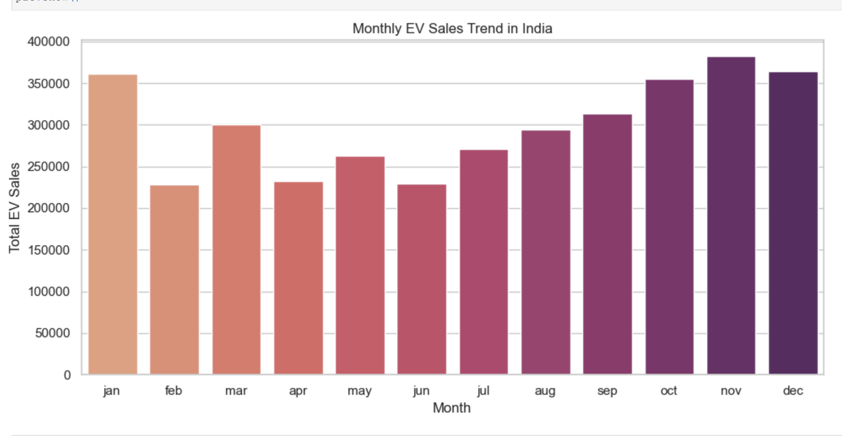


In [16]:
```
# Order of months
month_order = ['jan', 'feb', 'mar', 'apr', 'may', 'jun',
               'jul', 'aug', 'sep', 'oct', 'nov', 'dec']

monthly_sales = df.groupby('Month_Name', observed=True)['EV_Sales_Quantity'].sum().reindex(month_order)

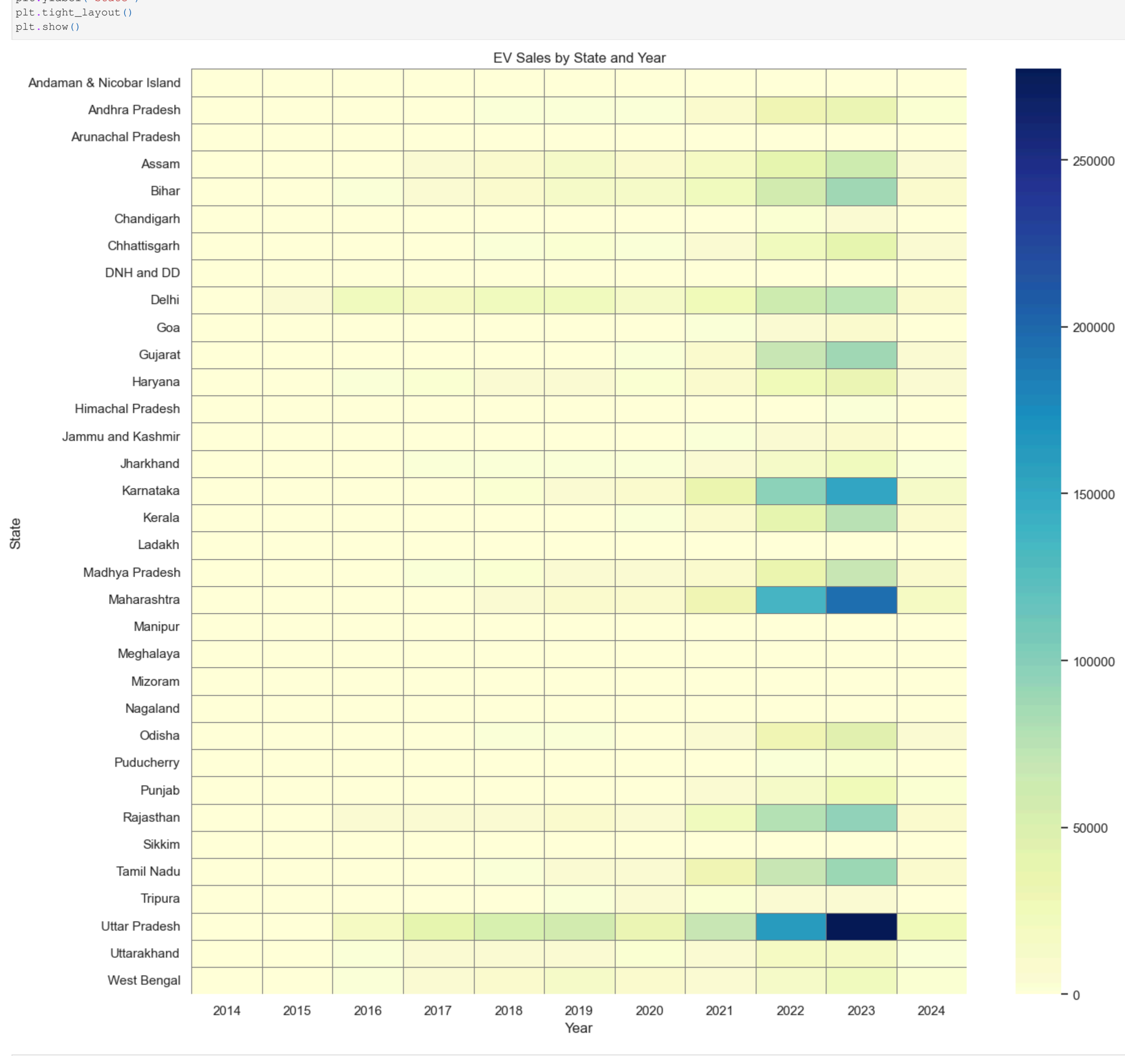
monthly_sales_df = monthly_sales.reset_index()
monthly_sales_df.columns = ['Month_Name', 'EV_Sales_Quantity']

plt.figure(figsize=(10, 5))
sns.barplot(
    data=monthly_sales_df,
    x='Month_Name',
    y='EV_Sales_Quantity',
    hue='Month_Name',
    palette='flare',
    dodge=False,
    legend=False
)
plt.title('Monthly EV Sales Trend in India')
plt.xlabel('Month')
plt.ylabel('Total EV Sales')
plt.tight_layout()
plt.show()
```



In [21]:
```
heatmap_data = df.pivot_table(
    values='EV_Sales_Quantity',
    index='State',
    columns='Year',
    aggfunc='sum',
    fill_value=0,
    observed=True
    )

plt.figure(figsize=(14, 12))
sns.heatmap(
    heatmap_data,
    cmap='YlGnBu',
    linewidths=0.5,
    linecolor='gray'
)
plt.title('EV Sales by State and Year')
plt.xlabel('Year')
plt.ylabel('State')
plt.tight_layout()
plt.show()
```



In [29]:
```
# top 10 states by total EV sales
top_10_states = df.groupby('State', observed=True)['EV_Sales_Quantity'].sum().sort_values(ascending=False).head(10).index

df_top_states = df[df['State'].isin(top_10_states)]

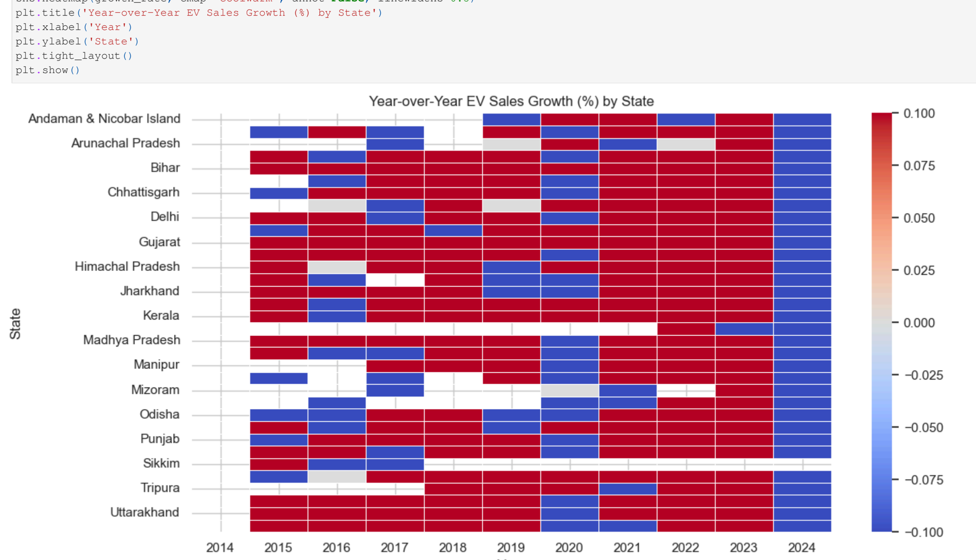
heatmap_data_top = df_top_states.pivot_table(
    values='EV_Sales_Quantity',
    index='State',
    columns='Year',
    aggfunc='sum',
    fill_value=0,
    observed=True
    )

plt.figure(figsize=(10, 6))
sns.heatmap(
    heatmap_data_top,
    cmap='YlOrRd',
    linewidths=0.5,
    linecolor='gray',
    annot=True,
    fmt='.0f'
)
plt.title('EV Sales by Year (Top 10 States Only)')
plt.xlabel('Year')
plt.ylabel('Top 10 States')
plt.tight_layout()
plt.show()
```



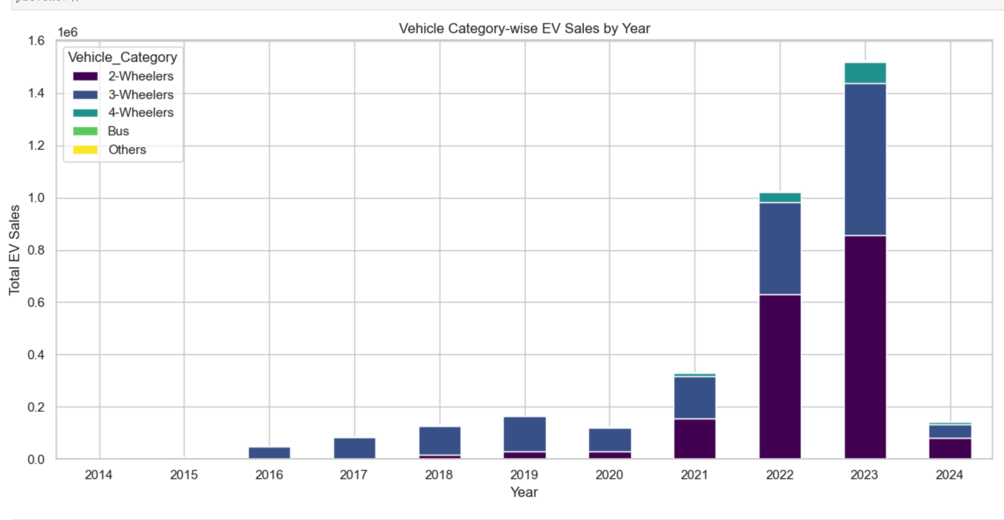
In [31]:
```
state_year = df.groupby(['State', 'Year'],observed=True)['EV_Sales_Quantity'].sum().unstack().fillna(0)
growth_rate = state_year.pct_change(axis=1) * 100  # YoY %

plt.figure(figsize=(12, 6))
sns.heatmap(growth_rate, cmap='coolwarm', annot=False, linewidths=0.5)
plt.title('Year-over-Year EV Sales Growth (%) by State')
plt.xlabel('Year')
plt.ylabel('State')
plt.tight_layout()
plt.show()
```



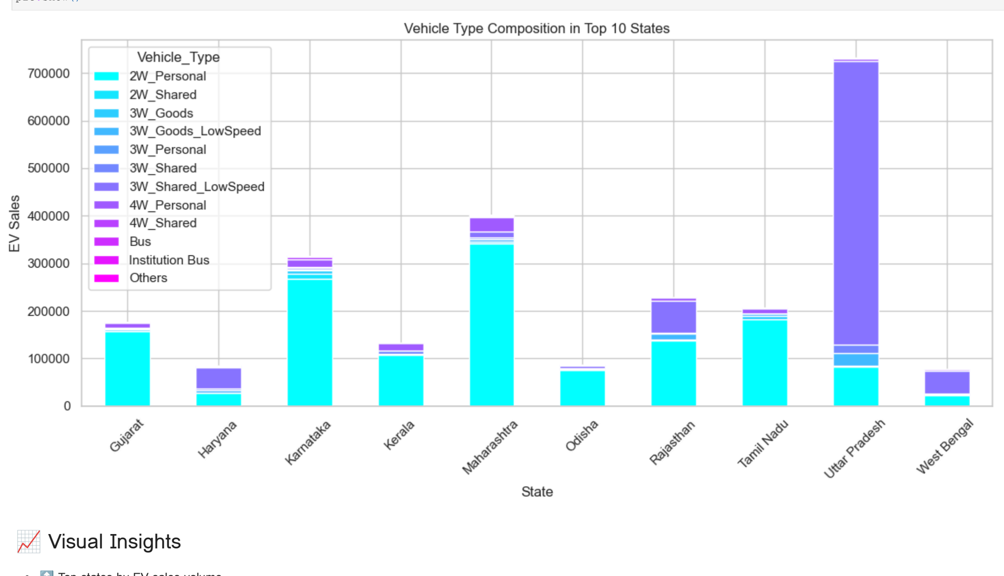
In [34]:
```
category_year = df.groupby(['Year', 'Vehicle_Category'],observed=True)['EV_Sales_Quantity'].sum().unstack().fillna(0)

category_year.plot(kind='bar', stacked=True, colormap='viridis', figsize=(12, 6))
plt.title('Vehicle Category-wise EV Sales by Year')
plt.ylabel('Total EV Sales')
plt.xlabel('Year')
plt.xticks(rotation=0)
plt.tight_layout()
plt.show()
```



In [35]:
```
top_states = df['State'].value_counts().head(10).index
state_type = df[df['State'].isin(top_states)].groupby(['State', 'Vehicle_Type'],observed=True)['EV_Sales_Quantity'].sum().unstack().fillna(0)

state_type.plot(kind='bar', stacked=True, colormap='cool', figsize=(12, 6))
plt.title('Vehicle Type Composition in Top 10 States')
plt.ylabel('EV Sales')
plt.xlabel('State')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



📈 Visual Insights

- 🔝 Top states by EV sales volume
- 🛵 Vehicle category dominance (2W, 3W, 4W)
- 📅 Monthly/seasonal trends in EV adoption
- 🗺️ Heatmap of state-wise growth year over year
- 📊 Cumulative EV sales curves for top states
- 📈 YoY growth percentage heatmap

📌 Conclusion

- EV sales in India have grown consistently from 2014 to 2023, with massive spikes post-2020.
- Uttar Pradesh, Maharashtra, and Karnataka lead in total EV sales volume.
- Two-wheelers (2W) dominate the market, but three-wheelers (3W) show strong regional growth.
- Monthly trends show steady year-round sales, with mild surges around mid-year.
- Some states like Assam and Bihar have shown impressive YoY growth in recent years.